Chapter 11. Universal Approximation and Barron's Theorem

Lecture: Oct. 21 (UAT: measures, normed spaces, Cybenko's theorem; Barron class)

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1 Three Sources of Error

Let f^* be the ideal target, $\mathcal{F}$ the class of functions a given architecture can represent, f_{app} the best element of $\mathcal{F}$, f_{gen} the best element on the finite training sample (empirical risk minimiser) and f_{opt} what the optimiser actually returns. The total error splits as

$$\underbrace{f^* \to f_{\text{app}}}_{\text{approximation}} + \underbrace{f_{\text{app}} \to f_{\text{gen}}}_{\text{generalisation (estimation)}} + \underbrace{f_{\text{gen}} \to f_{\text{opt}}}_{\text{optimisation}}.$$

Chapters 7–8 dealt with optimisation; this chapter shows that the approximation error of one-hidden-layer networks can be made arbitrarily small (universality) and, for Barron functions, quantifies it.

2 Preliminaries from Analysis

2.1 Measures

Definition: measures on a compact set $K \subset \mathbb{R}^n$

The **Borel sets** of K form the smallest collection containing all open subsets of K and closed under complements in K and countable unions. A (nonnegative) **measure** μ assigns $\mu(A) \in [0, \infty]$ to Borel sets with

$$\mu(\emptyset) = 0, \qquad \mu\Big(\bigcup_{j=1}^{\infty} A_j\Big) = \sum_{j=1}^{\infty} \mu(A_j) \quad \text{for pairwise disjoint } A_j.$$

A **signed measure** is the same with real (possibly negative) values, e.g. a difference of two finite measures. $\mu = 0$ means $\mu(A) = 0$ for every Borel set A. $M(I_n)$ denotes the finite signed regular Borel measures on the unit cube $I_n = [0,1]^n$.

2.2 Normed spaces and density

Definition: linear and normed spaces

A real **linear space** is a set closed under addition and multiplication by reals obeying the usual vector-space rules; e.g. $C(K) = \{f : K \to \mathbb{R} \text{ continuous}\}$, since $af + bg \in C(K)$ with $(af + bg)(x) = af(x) + bg(x)$. A **normed linear space** is a linear space X with $\|\cdot\| : X \to [0, \infty)$ such that

$$\|u\| = 0 \iff u = 0, \qquad \|cu\| = |c|\|u\|, \qquad \|u + v\| \le \|u\| + \|v\|.$$

On $C(K)$ we use the sup norm $\|f\|_\infty = \max_{x \in K} |f(x)|$. A subset $D \subseteq X$ is **dense** in X if for every $f \in X$ and $\varepsilon > 0$ there is $g \in D$ with $\|f - g\| < \varepsilon$.

Density in $(C(I_n), \|\cdot\|_\infty)$ is exactly uniform approximation: every continuous function can be approximated within ε at every point simultaneously.

3 Cybenko's Universal Approximation Theorem

Definition: sigmoidal and discriminatory

$\sigma : \mathbb{R} \to [0,1]$ is **sigmoidal** if $\lim_{t\to-\infty} \sigma(t) = 0$ and $\lim_{t\to+\infty} \sigma(t) = 1$. For $\mu \in M(I_n)$, σ is **discriminatory** if

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \quad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R} \qquad \Longrightarrow \qquad \mu = 0.$$

Theorem: Cybenko 1989 (UAT)

Let σ be any continuous sigmoidal function. Then finite sums

$$G(\mathbf{x}) = \sum_{j=1}^{N} \alpha_j\, \sigma(\mathbf{w}_j^\top \mathbf{x} + \theta_j), \qquad \mathbf{w}_j \in \mathbb{R}^n,\ \alpha_j, \theta_j \in \mathbb{R},\ N \in \mathbb{N},$$

are dense in $C(I_n)$: for every continuous f and $\varepsilon > 0$ some G satisfies $\sup_{\mathbf{x}\in I_n} |f(\mathbf{x}) - G(\mathbf{x})| < \varepsilon$.

The proof combines three ingredients: a Hahn–Banach consequence (Lemma 3), the Riesz representation theorem (Fact), and the discriminatory property (Lemma 1).

3.1 The functional-analytic lemmas

Theorem: Lemma 3 (Hahn–Banach)

Let W be a linear, non-dense subspace of a normed linear space X. Then there is a bounded linear functional L on X with $L \neq 0$ on X and $L|_W = 0$.

Proof (supplement; skipped in lecture). The closure $\overline{W}$ is a closed subspace and $\overline{W} \neq X$, so pick $x_0 \notin \overline{W}$; then $d = \inf_{w\in\overline{W}} \|x_0 - w\| > 0$. On $\overline{W} \oplus \mathbb{R}x_0$ define $\ell(w + tx_0) = t$. It is linear, vanishes on W, $\ell(x_0) = 1$, and bounded: $|t|d \le |t|\|x_0 + w/t\| = \|w + tx_0\|$ for $t \neq 0$, so $\|\ell\| \le 1/d$. The Hahn–Banach theorem extends ℓ to X with the same norm. ■

Theorem: Fact (Riesz representation theorem)

Let $K \subset \mathbb{R}^n$ be compact. For every bounded linear functional F on $C(K)$ there is a unique finite signed (regular Borel) measure μ on K with

$$F(f) = \int_K f(\mathbf{x})\, d\mu(\mathbf{x}) \qquad \text{for all } f \in C(K).$$

(For positive functionals the measure is nonnegative.)

Theorem: Lemma 4

Let W be a linear, non-dense subspace of $C(I_n)$. Then there is $\mu \in M(I_n)$, $\mu \neq 0$, with $\int_{I_n} h\, d\mu = 0$ for all $h \in W$.

Proof. Lemma 3 with $X = C(I_n)$ gives a bounded linear $L \neq 0$ with $L|_W = 0$. Riesz represents $L(f) = \int f\, d\mu$; $\mu \neq 0$ because $L \neq 0$; and for $h \in W$, $\int h\, d\mu = L(h) = 0$. ■

3.2 Density for discriminatory functions

Theorem: Lemma 5

If σ is continuous and discriminatory, the sums G above are dense in $C(I_n)$.

Proof. Since σ is continuous, $W = \{G\}$ is a linear subspace of $C(I_n)$ (sums and scalar multiples of such sums are again such sums). Suppose W is not dense. By Lemma 4 there is $\mu \neq 0$ with $\int h\, d\mu = 0$ for all $h \in W$, in particular

$$\int_{I_n} \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu(\mathbf{x}) = 0 \qquad \text{for all } \mathbf{w} \in \mathbb{R}^n,\ \theta \in \mathbb{R}.$$

Because σ is discriminatory, $\mu = 0$: a contradiction. ■

3.3 Continuous sigmoidals are discriminatory

Theorem: Lemma 1

Every continuous sigmoidal function is discriminatory for all $\mu \in M(I_n)$.

Proof (supplement; skipped in lecture). Assume $\int \sigma(\mathbf{w}^\top \mathbf{x} + \theta)\, d\mu = 0$ for all $\mathbf{w}, \theta$. Fix $\mathbf{w}, \theta, \varphi$ and let $\sigma_\lambda(\mathbf{x}) = \sigma(\lambda(\mathbf{w}^\top \mathbf{x} + \theta) + \varphi)$. As $\lambda \to \infty$, $\sigma_\lambda \to 1$ on the open half-space $\Pi = \{\mathbf{w}^\top \mathbf{x} + \theta > 0\}$, $\to 0$ on $\{\mathbf{w}^\top \mathbf{x} + \theta < 0\}$ and $= \sigma(\varphi)$ on the hyperplane $H = \{\mathbf{w}^\top \mathbf{x} + \theta = 0\}$. The σ_λ are bounded by 1, so dominated convergence gives

$$0 = \int \sigma_\lambda\, d\mu \ \longrightarrow\ \mu(\Pi) + \sigma(\varphi)\mu(H) \qquad \text{for every } \varphi.$$

Letting $\varphi \to -\infty$ and $\varphi \to +\infty$ yields $\mu(\Pi) = 0$ and $\mu(H) = 0$ for all half-spaces and hyperplanes. Fix $\mathbf{w}$ and define $F(h) = \int h(\mathbf{w}^\top \mathbf{x})\, d\mu(\mathbf{x})$ for bounded measurable $h : \mathbb{R} \to \mathbb{R}$. F is linear and bounded, and $F(\mathbf{1}_{[\theta,\infty)}) = \mu(\Pi) + \mu(H) = 0$, $F(\mathbf{1}_{(\theta,\infty)}) = 0$, hence F vanishes on indicators of intervals, on simple functions, and by uniform approximation on all bounded measurable h. Taking $h = \cos$ and $h = \sin$ gives $\int e^{i\mathbf{w}^\top \mathbf{x}}\, d\mu(\mathbf{x}) = 0$ for every $\mathbf{w}$: the Fourier transform of μ vanishes, so $\mu = 0$. ■

Proof of Cybenko's theorem. Lemma 1 shows σ is discriminatory; Lemma 5 gives density. ■

Remark: what the theorem says and does not say

One hidden layer with enough neurons approximates any continuous function on a compact set uniformly (the cube can be replaced by any compact K after rescaling). The theorem is an existence result: it gives no bound on N, no algorithm to find the weights, and no statement about generalisation. Later work extends it to every continuous non-polynomial activation, including ReLU (Leshno et al. 1993). Intuition in 1D: a steep sigmoid is almost a step, a difference of two steps is a bump, and a sum of bumps approximates any continuous function.

4 Barron's Theorem: a Dimension-Free Rate

4.1 Fourier preliminaries

For $f \in L^1(\mathbb{R}^d)$ the Fourier transform is $\hat{f}(\boldsymbol{\xi}) = \int_{\mathbb{R}^d} f(\mathbf{x}) e^{-2\pi i \boldsymbol{\xi}^\top \mathbf{x}}\, d\mathbf{x}$. The Sobolev space $H^s(\mathbb{R}^d)$ consists of f with $\int |\hat{f}(\boldsymbol{\xi})|^2 (1 + |\boldsymbol{\xi}|^2)^s\, d\boldsymbol{\xi} < \infty$.

Definition: Barron class

For $C > 0$,

$$\Gamma_C = \Big\{ f \in L^1(\mathbb{R}^d) : \ \|\hat{f}\|_{L^1(\mathbb{R}^d)} < \infty, \ \int_{\mathbb{R}^d} |2\pi\boldsymbol{\xi}| \, |\hat{f}(\boldsymbol{\xi})| \, d\boldsymbol{\xi} < C \Big\}.$$

On a domain Ω one also uses the spectral Barron norm $\|f\|_{\mathcal{B}^s(\Omega)} = \inf_{f_e|_\Omega = f} \int_{\mathbb{R}^d} (1 + |\boldsymbol{\xi}|)^s |\hat{f}_e(\boldsymbol{\xi})| \, d\boldsymbol{\xi}$, the infimum over all extensions f_e of f beyond Ω.

Theorem: Barron 1993

Let σ be sigmoidal, $f \in \Gamma_C$ and $B_1^d = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| \le 1\}$. Then for every $c > 4C^2$ and every $N \in \mathbb{N}$ there is a network f_N with N neurons such that

$$\frac{1}{|B_1^d|} \int_{B_1^d} |f(\mathbf{x}) - f_N(\mathbf{x})|^2 \, d\mathbf{x} < \frac{c}{N}.$$

The rate $O(1/N)$ does not depend on the dimension d; for generic smooth functions (e.g. approximation by fixed bases) the number of terms needed typically grows exponentially in d. The price is the assumption $f \in \Gamma_C$, a condition on the Fourier spectrum.

4.2 Proof idea: Maurey's empirical method

Theorem: approximation of convex combinations

Let $\mathcal{H}$ be a Hilbert space and $\mathcal{G} \subset \mathcal{H}$ with $\|g\| \le b$ for all $g \in \mathcal{G}$. If f is in the closed convex hull of $\mathcal{G}$, then for every N there are $g_1, \dots, g_N \in \mathcal{G}$ with

$$\Big\| f - \frac{1}{N} \sum_{k=1}^{N} g_k \Big\|^2 \le \frac{b^2 - \|f\|^2}{N} + \varepsilon \quad (\text{any } \varepsilon > 0).$$

Proof. First let $f = \sum_m \lambda_m h_m$ be a finite convex combination. Draw $g_1, \dots, g_N$ iid with $P(g_k = h_m) = \lambda_m$; then $\mathbb{E} g_k = f$ and, by independence,

$$\mathbb{E}\Big\| f - \frac{1}{N} \sum_{k} g_k \Big\|^2 = \frac{1}{N} \mathbb{E}\|g_1 - f\|^2 = \frac{1}{N} (\mathbb{E}\|g_1\|^2 - \|f\|^2) \le \frac{b^2 - \|f\|^2}{N}.$$

A random variable is not always above its mean, so some realisation achieves the bound. For f in the closure, approximate it by a finite convex combination first. ∎

Barron writes $f(\mathbf{x}) - f(\mathbf{0})$, via Fourier inversion, as an average over frequencies of functions $\cos(2\pi\boldsymbol{\xi}^\top \mathbf{x} + \phi) - \cos\phi$, each of which is a limit of combinations of sigmoidal step functions $\sigma(\mathbf{w}^\top \mathbf{x} + \theta)$; the Barron norm bounds the total mass, so $f - f(\mathbf{0})$ lies in the closed convex hull of a set of functions bounded by $2C$ in L^2 of the normalised ball. Maurey's lemma with $b = 2C$ gives $4C^2/N$; any $c > 4C^2$ absorbs the ε.